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F1 Real-Time Strategy

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Abstract

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1 Introduction

1.1. Context and Motivation

Template intent. Explain why Formula 1 pit-wall decision support is a strong case for streaming analytics plus machine learning.

- Start from the operational problem: decisions are sequential, time-critical, and made under uncertainty.
- Explain why batch-only analysis is insufficient for race-time decisions.
- Position event-time processing and low-latency architecture as enabling conditions.

Suggested references: [1, 8, 20, 31].

1.2. Research Scope and Objectives

Template intent. Define exactly what the thesis will and will not do.

- Scope: pit-stop recommendation as a supervised lap-level decision task in a streaming architecture.
- Comparison target: deterministic heuristic strategy engine vs learned model on the same replay stream.
- Explicit non-goals: full race-winning simulation, proprietary telemetry replication, and team-specific private strategy rules.
- Add 2–4 research questions, each measurable in Chapter 6.

Suggested references: [13, 19, 28, 29].

1.3. Contributions

Template intent. List the contribution claims in a thesis-safe and testable format.

- A reproducible event-time data engineering pipeline for F1 replay analytics.
- A leakage-aware ML protocol for pit-stop recommendation.
- A fair comparison protocol between heuristic and ML decisions in identical operating conditions.
- A reproducibility package (data contracts, scripts, environment, and experiment documentation).

Suggested references: [9, 23, 25].

1.4. Thesis Organization

Template intent. Provide a concise chapter roadmap.

- Chapter 2: state of the art and research gap.
- Chapter 3: problem definition, data, labels, and evaluation metrics.
- Chapter 4: solution design and comparison protocol.
- Chapter 5: implementation choices and reproducibility.
- Chapter 6: evaluation campaign and discussion.
- Chapter 7: conclusions, limitations, and future work.

2 State of the Art

2.1. Streaming Data Analytics

Template intent. Summarize stream-processing principles relevant to this thesis.

- Define unbounded data, event time, out-of-order events, watermarks, and correctness vs latency trade-off.
- Explain why a replayed historical race can still be processed as an event stream.
- Connect system guarantees to downstream ML reliability and reproducibility.

Suggested references: [1, 8, 20].

2.2. Complex Event Processing

Template intent. Explain CEP as the bridge between raw streams and tactical race signals.

- Define patterns, windows, sequence constraints, and event-time matching.
- Motivate CEP for lift-and-coast, tire-drop, and other race-state transitions.
- Clarify how CEP outputs differ from ML outputs and why both are needed in your architecture.

Suggested references: [1, 8, 31].

2.3. Platforms for Stream and Event Processing

2.3.1. Apache Kafka

Template intent. Describe Kafka role in your system.

- Topic partitioning, ordering assumptions, replayability, and decoupling of producers/consumers.

- Why replay-by-offset is essential for deterministic audits and ML dataset regeneration.

Suggested references: [20].

2.3.2. Apache Flink

Template intent. Describe Flink role in your system.

- Event-time operators, stateful processing, checkpointing, and fault tolerance.
- Why Flink is suitable for both online alerting and offline reproducible extraction.

Suggested references: [1, 8].

2.3.3. Related Technologies and Trade-offs

Template intent. Briefly compare alternatives and justify your stack.

- Compare latency and semantics trade-offs at a high level.
- State why your selected stack best matches thesis goals: correctness, replayability, and integration simplicity.

Suggested references: [31].

2.4. Machine Learning for Motorsport and Strategy Optimization

Template intent. Organize literature by paradigm, then by methodology quality.

- Supervised classification paradigm for pit-stop timing.
- Optimization/planning paradigm for tire and stop sequencing.
- Reinforcement learning paradigm for sequential control.
- Explicitly separate domain results from methodology quality (splits, metrics, calibration, significance).

Suggested references: [9, 13, 19, 23, 25, 28, 29].

Methodology anchors to cite in this section: [2–6, 10–12, 14–17, 21, 22, 24, 26, 27, 30].

2.5. Applications in Formula 1 Analytics

Template intent. Focus on direct F1 evidence and identify operational gaps.

- What has been done with public F1 data (features, targets, models, evaluation).
- What is still missing (streaming deployment, fair heuristic-vs-ML comparison, robust validation design).
- Position tire degradation modeling as supporting signal for pit-stop decisions.

Suggested references: [7, 13, 18, 19, 28].

2.6. Summary and Research Gaps

Template intent. Close Chapter 2 with a short, explicit gap list that leads into Chapter 3.

- Gap 1: Most published F1 ML evidence is batch-oriented.
- Gap 2: Validation and leakage controls are often weak or under-reported.
- Gap 3: Probability quality and statistical significance are rarely central.
- Gap 4: Direct heuristic-vs-ML comparisons in a streaming setting are limited.

3 | Problem Statement and Data

3.1. Formula 1 Pit Stop Decision Problem

Template intent. Define the prediction task unambiguously.

- Inference unit: driver-lap decision point.
- Target: binary label, pit in next K laps (state your fixed K and rationale).
- Operational interpretation: decision support signal, not autonomous control.

Suggested references: [9, 13, 19, 28, 29].

3.2. Data Sources and Acquisition

3.2.1. FastF1 Data Interface

Template intent. Explain source provenance and reproducibility constraints.

- Seasons/races included, extraction process, and versioning policy.
- Data quality caveats (missing telemetry, weather noise, incidents).

Suggested references: [13, 19, 28].

3.2.2. Streaming Inputs and Event Types

Template intent. Define each stream as a contract.

- Telemetry stream, lap stream, and track-status stream.
- Event-time semantics, ordering assumptions, and replay settings.

Suggested references: [1, 8, 20].

3.3. Dataset Description

3.3.1. Telemetry Stream

Template intent. List fields and their modeling role in few words.

- Mention sampling rate, key signals, and normalization choices.
- State known limitations and filtering rules.

3.3.2. Lap Stream

Template intent. Describe lap-level context and pace variables.

- Timing, stint, compound, tire age, and gap-derived features.
- Explain why lap-level aggregation is aligned with pit decision timing.

3.3.3. Track Status Stream

Template intent. Describe race-neutralization context.

- Encoding of green/yellow/VSC/SC and time availability constraints.
- How this stream is used for filtering, labeling, and stress tests.

3.4. Target Definition and Labeling Strategy

Template intent. Make labeling auditable and leakage-safe.

- Exact positive/negative label rule and boundary conditions.
- Inclusion/exclusion policy (for example wet phases, early laps, unresolved events).
- Explicit anti-leakage policy: no derived heuristic outputs as ML inputs.

Implemented fallback guardrail for missing post-pit gaps. The ground-truth operator now uses a conservative salvage rule only for same-rival cycles where direct post-pit gap reconstruction is unavailable. The fallback compares pre-pit and post-pit relative pace against the same rival using local green, non-pit lap windows with minimum sample constraints. If this evidence is not available, the cycle remains unresolved as `UNRESOLVED_MISSING_POST_GAP`. This preserves auditable uncertainty instead of forcing a strategic label from incomplete context, while staying aligned with lap-time decomposition

and race-time optimization reasoning from motorsport strategy literature [9, 18, 25].

3.4.1. Construct Validity and Label Scope

No official open Formula 1 pit-strategy ground-truth dataset was identified in the surveyed literature used in this thesis. As a result, labels are defined operationally from telemetry and lap context using explicit decision rules, not imported from an external authoritative source. This is consistent with recent motorsport studies that also operationalize pit outcomes from observable timing and position context [9, 13, 19, 25, 28].

The construct represented by the label is strategic pit outcome under observable race conditions, not total team performance. Therefore, the labeling scope intentionally excludes unobservable or weakly observable factors such as pit crew execution quality, hidden car damage, and team-internal strategy constraints. The unresolved policy is conservative by design: when rival context or post-pit comparability is insufficient, the event is retained in data lake artifacts but excluded from resolved binary supervision. This abstention policy limits label noise and leakage from ambiguous contexts [6, 26].

External validity is bounded to the seasons, regulations, and telemetry quality covered by the replayed dataset. Claims are therefore framed as construct-valid for this operational setting, with qualitative triangulation in Chapter 6 used to assess plausibility against known race narratives.

Suggested references: [6, 9, 18, 19, 25, 26, 28].

3.5. Evaluation Metrics

Template intent. Define metrics before results to avoid metric shopping.

- Primary discrimination metrics for imbalance: precision, recall, F1, AUC-PR.
- Calibration metrics: Brier score and reliability analysis.
- Statistical comparison protocol: paired significance tests with corrected interpretation.

Suggested references: [5, 10–12, 17, 21, 22, 24, 27, 30].

4 Solution and Design

4.1. System Architecture Overview

Template intent. Describe end-to-end architecture in one page.

- Producer, broker, stream processor, data lake, and dashboard roles.
- Explain how architecture choices support reproducibility and fair comparisons.

Suggested references: [1, 8, 20].

4.2. Dataflow and Processing Topology

Template intent. Specify operator order and data dependencies.

- Event-time path from ingestion to features/alerts/labels.
- Stateful operators and where context enters the pipeline.
- Contracts for sinks used by audit and ML.

4.2.1. Dual-Consumer Streaming Contract (Flink + ML Placeholder)

Template intent. Reserve this subsection for deployment-consistency design notes.

- State contract: Flink computes stateful lap-context features once and emits the same feature payload to offline sink and online topic (`f1-ml-features`).
- Document event-time ordering and watermark assumptions used to keep feature timestamps point-in-time consistent across replay and live serving.
- Describe Kafka decoupling between heuristic processing and ML inference consumer to preserve independent runtime behavior under the same telemetry stream.

- Keep raw-telemetry serving as a deferred ablation track after consistency and fairness gates are validated.

Suggested references: [1, 8, 20, 31].

4.3. Heuristic Strategy Engine Design

Template intent. Explain the handcrafted strategy logic at a conceptual level.

- Inputs, decision factors, and prioritization logic.
- Expected strengths and known failure modes.
- Clarify this module as baseline and operational comparator.

4.3.1. Pit-Cycle Label Resolution (Skeleton Notes)

Template intent. Keep this subsection structural and implementation-neutral in early drafts.

- Document fallback hierarchy order: direct rival-gap path, near-lap recovery, pace-shift fallback, unresolved abstention.
- Explain why conservative unresolved outcomes are preserved when evidence remains incomplete.
- Link rationale to lap-time decomposition and online optimization framing.

Suggested references: [9, 18, 25].

4.3.2. Timing-Pressure Actionability Alignment (Skeleton Notes)

Template intent. Record purpose and guardrails of timing-aware suggestion logic without locking final numeric detail.

- Summarize timing-pressure signals (pace trend, gap trend, strategy deficit, rival reaction, traffic context).
- Note green-condition actionability gating that downgrades weak early actionable alerts to monitoring.
- Capture near-window re-emission intent when timing pressure materially increases.
- Emphasize deterministic bounded behavior to keep the SDE baseline stable and auditable.

Suggested references: [7, 9, 18, 25, 31].

4.4. Machine Learning Formulation

Template intent. Specify model-family choices and data protocol without reporting outcomes yet.

- Candidate baselines (for example logistic/XGBoost and optional sequence model).
- Feature groups, leakage controls, and temporal/grouped split strategy.
- Calibration and thresholding plan for decision outputs.

Suggested references: [6, 13, 17, 19, 21, 24, 26, 28].

4.5. Comparison Protocol, Heuristic vs ML

Template intent. Define a fairness protocol before experiments.

- Same replay windows, same data availability assumptions, same excluded contexts.
- Use one shared feature contract for training and first live serving comparison to reduce training-serving skew.
- Report both predictive quality and operational behavior.
- Predefine significance testing and uncertainty reporting.

Suggested references: [11, 12, 22, 30].

5 | Implementation Experience

5.1. Engineering Stack and Environment

Template intent. Keep this section factual and reproducible.

- Report runtime stack, versions, infrastructure, and packaging.
- Document environment constraints and reproducibility notes.
- Describe what is deterministic vs what can vary between runs.

5.2. Pipeline Implementation

5.2.1. Ingestion and Replay Components

Template intent. Describe implementation decisions that affect data fidelity.

- Replay timing policy, topic schema, and serialization choices.
- Error handling and idempotency notes.

5.2.2. Flink Operators and Real-time Alerts

Template intent. Summarize operator responsibilities and state management.

- Per-operator responsibilities and key state variables.
- Watermarks/checkpointing and practical trade-offs encountered.

PitStopEvaluator evolution (skeleton).

- Note the fallback hierarchy and where pace-shift recovery is used.
- Record deterministic abstention behavior when post-pit evidence is incomplete.
- Link rationale to lap-time decomposition and online optimization references.

PitStrategyEvaluator evolution (skeleton).

- Record added timing-pressure context (pace and gap trends) at per-driver state level.
- Capture actionability gate intent: avoid broad early actionable alerts in green conditions.
- Emphasize deterministic bounded logic as a stable SDE baseline.

Replay-speed stress behavior (skeleton).

- Keep one concise note on 500x stress validation and season completion.
- Mention duplicate-key artifacts as operational side effects, not comparator semantic changes.
- State that comparator robustness relies on deterministic de-duplication before matching.

5.2.3. Data Lake and Audit Layer

Template intent. Explain output contracts and quality checks.

- Sink layout, partitioning logic, and schema consistency policy.
- Audit scripts and contract-validation process.

Comparator diagnostics extension (skeleton).

- Note added timing-distance diagnostics for excluded and matched rows.
- Explicitly state that matching semantics remain unchanged.
- Use this subsection later for compact attribution of horizon misses vs unresolved outcomes.

5.3. Model Training Pipeline

Template intent. Keep this broad until final experiments stabilize.

- Training data extraction path and feature assembly overview.
- Split strategy implementation and anti-leakage safeguards.
- Hyperparameter search, calibration step, and model artifact versioning.

Suggested references: [6, 17, 21, 24, 26].

5.4. Real-time Inference Integration

Template intent. Describe integration options and chosen approach without overcommitting to one architecture in early drafts.

- Integration pattern selected (in-engine, service, or decoupled consumer).
- Input contract, output topic/schema, and latency budget assumptions.
- Fallback and observability strategy for inference failures.

5.4.1. Training-Serving Consistency Path (Option A Skeleton)

Template intent. Keep this subsection as an implementation checklist for the first live ML comparison phase.

- Record one-feature-authority rule: Flink computes rolling/context features and emits both offline JSONL and online Kafka payloads.
- Note decoupled ML inference consumer flow: `f1-ml-features` input to `f1-ml-predictions` output, independent from heuristic topic path.
- Document serving bundle contract items (feature-column order, threshold metadata, model version tag).
- Reserve explicit checks for temporal leakage and train-serve transform drift.

Suggested references: [6, 8, 20, 26].

6 | Evaluation Campaign

6.1. Experimental Setup and Parameters

Template intent. Define the evaluation setting before showing any numbers.

- Races/seasons included, replay speed policy, and compute environment.
- Fixed parameters and tunable parameters.
- Repetition policy for robustness.

6.1.1. Replay-Speed Sensitivity Protocol

Template intent. Keep this subsection as a compact robustness checklist.

- Compare nominal and high-speed replay settings for operational stability.
- Check season representativeness, duplicate-key artifacts, and sink contract validity.
- Confirm comparator stability after deterministic de-duplication.

Suggested references: [1, 8, 31].

6.2. Dataset Split and Validation Protocol

Template intent. Explain exactly how leakage is prevented.

- Grouped/temporal split strategy and rationale.
- What is held out, when, and why.
- Stress-test subsets (for example SC/VSC/wet) vs core training subset.

Suggested references: [6, 26].

6.2.1. Training-Serving Consistency Audit Protocol

Template intent. Keep this subsection as a gate checklist before reporting live ML-vs-heuristic outcomes.

- Compare offline `ml_features` rows and live `f1-ml-features` payloads on race-driver-lap keys.
- Verify point-in-time constraints for each feature family used by serving inference.
- Report schema/feature parity rates before model-quality metrics.
- Report operational live-path metrics: prediction availability and latency ($p50$, $p95$).

Suggested references: [1, 6, 26, 31].

6.3. Baseline Definitions

6.3.1. Heuristic Baseline

Template intent. Specify heuristic decision rule version used in the comparison.

- Inputs available at inference time.
- Decision output format and thresholding policy.

6.3.2. Machine Learning Baseline(s)

Template intent. Declare model variants and calibration/threshold steps.

- Model families and selected final candidates.
- Calibration method and operating-point selection logic.

Suggested references: [5, 10, 17, 21, 24, 27].

6.4. Results on Recommendation Quality

Template intent. Add final tables/figures only when experiments are frozen.

- Report precision, recall, F1, AUC-PR, and calibration metrics.
- Provide race-level and aggregate-level views.
- Include uncertainty intervals and significance tests.

Suggested references: [11, 12, 22, 30].

6.4.1. Current SDE Gate Status After Timing-Pressure Pass

Template intent. Keep one concise locked-baseline snapshot for later ML comparison.

- Fixed-rule SDE baseline at $H = 2$: scored rows 113, precision 0.6903, TP 78, FP 35.
- Mark this as the frozen comparator reference before ML transition.
- Record high-level residual pattern: horizon timing misses dominate, unresolved post-gap remains secondary.

6.5. Latency and Throughput Results

Template intent. Keep this section focused on operational feasibility.

- End-to-end latency distribution and throughput under replay loads.
- Compare overhead introduced by ML inference path vs heuristic-only path.

6.6. Error Analysis and Failure Modes

Template intent. Explain where and why decisions fail.

- Analyze errors by race context (track, compound, neutralization regime).
- Distinguish data issues, model limitations, and unavoidable ambiguity.

6.6.1. Qualitative Spot-Check Protocol for Label Plausibility

Template intent. Keep this as a reproducible qualitative-audit placeholder.

- Define a stratified race/event sample for manual plausibility checks.
- Record context, assigned class, resolution path, and narrative consistency.
- Report agreement separately for resolved labels and unresolved rationale.

Suggested references: [6, 19, 26, 28].

6.6.2. Robustness of the Conservative Unresolved Policy

Template intent. Keep this subsection focused on coverage-vs-purity trade-off evidence.

- Report unresolved distributions by reason code.
- Quantify comparator exclusions attributable to unresolved outcomes.
- Reserve sensitivity discussion for conservative fallback paths only.
- Keep unresolved reduction as a secondary lever after timing alignment.

Suggested references: [6, 14, 26].

6.7. Discussion

Template intent. Synthesize findings without overclaiming.

- Interpret practical implications for pit-wall support.
- State what can be concluded confidently and what remains uncertain.

7 Conclusion

7.1. Obtained Results

Template intent. Keep this section high-level until final numbers are fixed.

- Summarize what was built and evaluated.
- Restate outcome at the level of evidence (not hype): where ML improved, matched, or underperformed heuristic logic.
- Mention both decision quality and operational feasibility.

7.2. Limitations

Template intent. Be explicit and concrete.

- Data limitations (coverage, label ambiguity, special race conditions).
- Method limitations (model class, assumptions, calibration drift).
- System limitations (latency budget, integration constraints, observability).

7.3. Future Work

Template intent. Propose realistic next steps tied to thesis findings.

- Drift-aware adaptation and online recalibration.
- Richer decision spaces (for example compound recommendation jointly with timing).
- Tighter integration patterns and broader season-level validation.
- Training-serving skew mitigation when telemetry is streamed in parallel to Flink and ML inference.

Suggested references: [2–4, 15, 16].

7.3.1. Deployment Consistency Placeholder (Flink + ML)

Template intent. Keep this subsection as a future-work marker until deployment experiments are run.

- Define how online features should be generated at serving time to match training assumptions.
- Specify consistency checks between offline Flink-derived training features and live serving features.
- Reserve latency and fallback requirements for the final system design chapter pass.

7.4. Overall Discussion

Template intent. Close by positioning your thesis against the three strategy paradigms.

- Optimization/planning paradigm.
- Reinforcement-learning paradigm.
- Supervised classification in streaming deployment.

Suggested references: [9, 13, 19, 25, 28, 29].

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A | Appendix A, Additional Experimental Material

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Variable	Description	Unit

Acknowledgements